Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 11: Transportation in Animals and Plants

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	11 — Transportation in Animals and Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors weaponise the classic flips — arteries-low-pressure, veins-have-valves-because-of-high-pressure, ventricles-receive/atria-pump, pulse-is-a-valve-snap, urea-is-useful-food or made-in-the-bladder, xylem-carries-food, phloem-only-downward, transpiration-stops-in-heat, wilting-means-poisoned-soil, and diffusion-suffices-for-large-bodies — but here they are buried inside multi-step data reads, ratio and rate computations, and cross-system synthesis so a half-remembered rule still lands on a wrong option. Do not write on this paper — mark answers on your answer sheet.

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1. A laboratory report on a patient lists three findings: oxygen-carrying capacity normal, blood clots in the usual time, but bacterial infections keep recurring and spreading. Which single blood component is most plausibly deficient, given that the other two functions are intact?

(A) White blood cells, since defence against germs is the only listed function that is failing
(B) Red blood cells, since recurring infection always means too little haemoglobin
(C) Platelets, because spreading infection is really a sign of poor clotting
(D) Plasma, because the watery part of blood is what destroys invading bacteria

2. Two tubes of blood are spun until the parts settle. Tube X separates into a tall clear pale-yellow upper column with a thin packed layer below; tube Y from a different person shows a much shorter yellow column and a thicker packed cell layer. Reading these two tubes, the person whose blood is Y has, relative to X:

(A) A larger proportion of plasma and fewer cells, because the yellow column is the cells

(B) More haemoglobin dissolved in the yellow upper layer, since plasma carries oxygen

(C) More platelets floating in the clear layer, which is why the layer is thinner

(D) A larger proportion of cells and a smaller proportion of the liquid plasma

3. Mature human red cells lose their nucleus as they form. A student claims this is a disadvantage. The best Class-7 evaluation of the trade-off is that losing the nucleus:

(A) Frees up internal space for more haemoglobin, improving oxygen carriage, even though the cell can no longer divide or repair itself

(B) Is purely harmful, since a cell with no nucleus cannot carry oxygen at all

(C) Lets the cell fight germs more aggressively in place of the white cells

(D) Allows the red cell to form blood clots when no platelets are present

4. A doctor explains anaemia and a clotting disorder to a family. In anaemia the person tires and is breathless; in the clotting disorder small cuts bleed a long time. Which pairing of cause to symptom is correct?

(A) Too little haemoglobin causes the breathlessness; too few platelets causes the slow clotting

(B) Too few platelets cause the breathlessness; too little haemoglobin causes the slow clotting

(C) Too few white cells cause both the breathlessness and the slow clotting

(D) Too much plasma causes the breathlessness; too many red cells cause the slow clotting

5. During a severe infection a child's white-cell count climbs steeply for several days, then falls back as the child recovers. The pattern is best explained by:

(A) The body producing extra defenders while germs are present and reducing them once the threat is cleared

(B) Red cells turning into white cells during illness and back into red cells on recovery

(C) Platelets multiplying to clot the infected blood and then dissolving away

(D) Plasma thickening into white cells and thinning again as the fever breaks

6. A nutrition poster says 'eat iron-rich food for strong blood.' A student must justify this in terms of function. The soundest justification is that dietary iron:

(A) Becomes the platelets, so without iron the blood cannot clot

(B) Forms the liquid plasma in which the blood cells float

(C) Builds the white cells that the body uses to fight germs

(D) Is built into haemoglobin, so without enough iron the blood carries less oxygen

7. Plasma carries the blood cells, but it also transports dissolved cargo around the body. Which combination of cargoes does plasma realistically carry in a Class-7 picture?

(A) Oxygen as its main load, because plasma is what binds oxygen

(B) Only the solid clot that forms at a wound

(C) Digested food, the waste urea, and other dissolved substances

(D) Whole undigested food and bone fragments for repair

8. A blood-bank technician says a transfusion was given specifically to restore a patient's failing oxygen supply to the tissues after heavy blood loss. The component of the donated blood most directly responsible for meeting that goal is the:

(A) Red cells, because their haemoglobin carries oxygen to the tissues

(B) Platelets, because oxygen delivery depends on fast clotting

(C) White cells, because fighting germs is how oxygen reaches tissues

(D) Plasma proteins, because the liquid alone supplies the oxygen

9. Bright cherry-red blood is drawn from one vessel and darker, duller red blood from another in the same healthy person. Within Class-7 science the most reasonable reading is that the brighter sample:

(A) Contains more platelets, which gives blood its brighter colour

(B) Is genuine blood while the dark sample is blood turning into urine

(C) Is carrying more oxygen bound to its haemoglobin than the darker sample

(D) Was blue inside the body and only reddened on contact with air

10. A single drop of blood under a microscope shows, by count, that one kind of cell vastly outnumbers the others, is small, round and lacks a nucleus. Identifying this most-numerous cell and matching it to its job gives:

(A) White cells, whose job is to clot the blood at wounds

(B) Platelets, whose job is to carry oxygen around the body

(C) Red cells, whose job is to carry oxygen

(D) Plasma cells, whose job is to fight invading germs

11. In an accident an artery and a vein are both cut. The artery throws blood out in strong spurts timed with the heartbeat, while the vein bleeds in a steady, low ooze. Synthesising vessel structure and the heart's action, the artery spurts because it:

(A) Carries blood pumped straight from the heart at high pressure that rises and falls with each beat

(B) Carries blood slowly back toward the heart, which makes it surge

(C) Has many valves that snap open and shut, producing the spurts

(D) Is thinner-walled than the vein, so blood escapes in bursts

12. A student measures wall thickness and counts valves in three vessels. Vessel 1: thick muscular wall, no valves. Vessel 2: thin wall, valves present. Vessel 3: wall one cell thick, no valves. Matching each to its name:

(A) 1 is a vein, 2 is an artery, 3 is a capillary(B) 1 is a capillary, 2 is an artery, 3 is a vein

(C) 1 is an artery, 2 is a capillary, 3 is a vein(D) 1 is an artery, 2 is a vein, 3 is a capillary

13. A classmate writes: 'Veins have valves because blood inside them moves at such high pressure that valves are needed to slow it down.' Correcting both the pressure and the purpose, the accurate statement is:

(A) Vein blood does move at very high pressure, and valves slow it as stated

(B) Veins have valves only to make the heartbeat sound louder

(C) Vein blood moves at low pressure, so valves are needed to stop it slipping backward and keep it heading to the heart

(D) Valves in veins actively pump the blood the way the heart does

14. Trace a red cell's vessel-type journey as it leaves the heart, passes through a leg muscle giving up oxygen, and returns to the heart. The correct order of vessel types is:

(A) Vein, then capillaries, then artery (B) Artery, then vein, then capillaries

(C) Capillaries, then artery, then vein (D) Artery, then capillaries, then vein

15. A student insists 'every artery carries oxygen-rich blood and every vein carries oxygen-poor blood, with no exceptions.' The most accurate correction names the exception:

(A) It is usually true, but the artery going to the lungs carries oxygen-poor blood and the vein returning from the lungs carries oxygen-rich blood

(B) It is completely true with no exceptions anywhere in the body

(C) It is completely false, since arteries always carry oxygen-poor blood

(D) It holds for adults only and reverses entirely in children

16. Comparing where pressure is highest and where flow is slowest in the circulation, the correct pairing is:

(A) Pressure is highest in the veins, and flow is slowest in the arteries

(B) Pressure is highest in the arteries, and flow is slowest in the capillaries

(C) Pressure is highest in the capillaries, and flow is slowest in the veins

(D) Pressure is highest and flow is slowest both in the arteries

17. Through pale forearm skin certain vessels look bluish, prompting a claim that the blood inside is blue. The scientifically correct account is that:

(A) The blood truly is blue and turns red only when it meets air at a cut

(B) The blood is dark red; it only looks bluish because of the way light passes through skin and vessel walls

(C) Only oxygen-poor blood is genuinely coloured blue inside the body

(D) The bluish look comes from blue plasma filling those vessels

18. Which single property belongs to arteries but is NOT typical of veins?

(A) Valves that prevent blood from flowing backward

(B) A wall that is only a single cell thick

(C) Thick, muscular, elastic walls built to take high pressure

(D) The job of carrying blood toward the heart

19. A capillary is one cell thick and threads between body cells. If, in a thought experiment, a capillary were rebuilt with thick muscular walls like an artery, the most direct consequence would be that:

(A) Exchange would speed up, because thicker walls push substances out faster
(B) Exchange of gases and substances between blood and body cells would be hindered, because the thin wall is what lets them pass
(C) The capillary would start pumping blood like a small heart
(D) Blood would clot inside it, since thick walls cause clotting

20. A vessel is described as carrying blood toward the heart, holding valves, and bleeding in a steady low-pressure ooze when cut. Naming the vessel and explaining the valves together gives:

(A) An artery, whose valves help it pump blood at high pressure
(B) A vein, whose valves keep low-pressure blood from slipping backward
(C) A capillary, whose valves slow exchange across its thin wall
(D) A vein, whose valves exist to withstand high pressure

21. The human heart keeps oxygen-rich and oxygen-poor blood on separate sides. A student asks what advantage this gives. The best answer is that keeping them unmixed:

(A) Lets fully oxygenated blood be sent to the body, so tissues get the most oxygen per unit of blood
(B) Makes the heartbeat sound louder through the stethoscope
(C) Lets the blood change colour faster as it passes through
(D) Allows the heart to beat without using any muscle at all

22. The two lower chambers of the heart have noticeably thicker muscular walls than the two upper chambers. Reading this structure as a clue to function, the thicker lower walls indicate that those chambers:

(A) Gently receive blood arriving from the veins, while the upper chambers pump it out
(B) Pump blood out of the heart with greater force, while the upper chambers mainly receive incoming blood
(C) Store urine briefly before it is passed out of the body
(D) Hold the valves that create the wrist pulse

23. A doctor listening with a stethoscope hears a repeated 'lub-dub.' Connecting the sound to its source, the 'lub-dub' is produced by:

(A) Blood being forced through the narrow capillaries
(B) The lungs drawing in and pushing out air
(C) The artery throbbing at the wrist as the pulse
(D) The heart's valves closing as blood is pumped through the chambers

24. A throb felt at the wrist matches the heartbeat one-for-one. Identifying what the throb actually is, the pulse is:

(A) Vein valves snapping shut in rhythm with the heart
(B) The blood clotting briefly at the skin with each beat

(C) The expansion of an artery each time the heart drives a surge of blood into it

(D) Air being pushed down the arm by the lungs

25. A nurse times a resting child and counts 27 pulse beats in 18 seconds. The pulse rate per minute is:

(A) 90 beats per minute
(B) 27 beats per minute
(C) 54 beats per minute
(D) 108 beats per minute

26. An athlete's heart beats 75 times per minute at rest. After a sprint it climbs to 150 beats per minute. Comparing the two states, in any given 20-second window after the sprint the heart pumps blood about:

(A) Twice as often as at rest, delivering oxygen and food faster and clearing waste quicker

(B) Half as often as at rest, to let the muscles cool down

(C) The same number of times, since rate does not change real flow

(D) Twice as often, but only to slow the blood and prevent clotting

27. Two resting adults are checked: one shows 70 beats per minute, the other 132. Given that a typical resting rate is about 72, the second adult's reading is best described as:

(A) Exactly the normal resting rate
(B) Lower than the usual resting rate
(C) Much higher than the usual resting rate
(D) Impossible for any living person

28. Following one heartbeat, oxygen-poor blood that returns from the body is sent to the lungs, and oxygen-rich blood from the lungs is sent on to the body. Reading this two-loop arrangement, its main benefit is that:

(A) The body receives oxygen-poor blood directly without visiting the lungs

(B) Blood is freshly loaded with oxygen at the lungs before each trip to the body, so tissues get oxygen-rich blood

(C) The heart can skip pumping to the lungs once oxygen is high

(D) Oxygen-rich and oxygen-poor blood are deliberately mixed for balance

29. A heart-rate trace shows resting beats every 0.8 seconds. Computing from this interval, the resting pulse rate is closest to:

(A) About 8 beats per minute
(B) About 48 beats per minute
(C) About 125 beats per minute
(D) About 75 beats per minute

30. A patient's heart is described as a muscular organ that contracts in a steady rhythm to move blood through the vessels. Picking the statement that is fully correct about the heart:

(A) It is a muscular organ that contracts rhythmically to pump blood through the vessels

(B) It is made mostly of bone that snaps shut to push blood along

(C) It filters nitrogen wastes out of the blood the way a kidney does

(D) It makes the body's food and sends it out through the phloem

31. A teacher draws an analogy: the heart in animals does a job that, in a plant raising water, is done by a different mechanism. The most reasonable matching is that the heart's pumping is paralleled in plants by:

(A) A tiny heart hidden in the roots that pumps water upward

(B) The transpiration pull that draws water up the xylem, although the plant has no actual pump

(C) The phloem squeezing water up to the leaves under pressure

(D) The stomata sucking water directly out of the soil

32. Excretion removes harmful wastes. Of the following body activities, the one that is NOT a form of excretion is:

(A) Taking in and digesting food in the gut(B) Removing urea from the blood as urine

(C) Losing water and salts through sweat

(D) Passing out dissolved nitrogen waste through the kidneys

33. Inside each kidney sit about a million microscopic filtering units that clean the blood. Naming these units and stating where they sit:

(A) Stomata, located inside the kidneys

(B) Nephrons, located inside the urinary bladder

(C) Nephrons, located inside the kidneys(D) Platelets, located inside the kidneys

34. Arrange the urinary structures in the exact order that newly formed urine travels through them on its way out of the body:

(A) Kidney, then urethra, then bladder, then ureter

(B) Bladder, then ureter, then kidney, then urethra

(C) Ureter, then kidney, then urethra, then bladder

(D) Kidney, then ureter, then urinary bladder, then urethra

35. Trace urea correctly from where it is produced to where it finally leaves the body:

(A) Produced in the kidneys, carried by the xylem, and passed out through stomata

(B) Produced in the body cells, carried by the blood, filtered out by the kidneys, and passed out in urine

(C) Produced in the bladder, carried by the arteries, and breathed out by the lungs

(D) Produced in the urine, carried by the ureters, and stored in the heart

36. A student writes that 'urea should be stored because it is a useful food the body saves for later.' The correct evaluation of this statement is that:

(A) It is right; urea is a stored food broken down when energy runs low

(B) It is wrong; urea is a toxic waste that would poison the body if allowed to build up

(C) It is right; urea is kept to colour the blood red

(D) It is partly right; urea is stored in the kidneys as a reserve

37. On a very hot day a footballer sweats heavily, while indoors on a cool day she sweats little but passes more urine. The kidneys and sweat glands together are best understood here as:

(A) Working against each other, so heavy sweating means the kidneys stop filtering blood

(B) Sharing the job of removing water and wastes, so when sweating rises the kidneys can let urine fall, and vice versa

(C) Doing unrelated jobs, since sweat is for cooling only and never removes waste

(D) Both taking in oxygen through the skin to replace what is lost in sweat

38. When both kidneys fail, harmful wastes accumulate in the blood, and a machine is used to clean it. Identifying the treatment and what it replaces:

(A) Transfusion, which takes over the kidneys' filtering job

(B) Transpiration, which is how a machine cleans human blood

(C) Circulation, a machine treatment that replaces the heart's pumping

(D) Dialysis, which takes over the kidneys' job of filtering wastes from the blood

39. Plants lack kidneys yet still deal with wastes. Which description of plant waste handling is correct?

(A) Plants release some waste through stomata and store some in parts like leaves, bark, gums, and resins

(B) Plants filter all their waste through a pair of plant kidneys

(C) Plants send every waste up the xylem and out through the roots

(D) Plants make no waste of any kind and so need no removal

40. Sticky gum oozing from a tree trunk and salts deposited in older leaves that later drop are both best interpreted as:

(A) Food the plant is deliberately sending down to its roots

(B) Water being pulled upward toward the leaves

(C) Ways a plant stores or releases its waste products

(D) Oxygen the plant releases through the bark at night

41. Comparing excretion in plants with excretion in animals, the most accurate Class-7 statement is that plant excretion is:

(A) Generally slower and simpler, since plants make less harmful waste and can store some of it

(B) Faster, because plants have many small kidneys

(C) Carried out entirely by a heart pumping waste through vessels

(D) Impossible, because plants produce no waste to remove

42. A potted plant is watered, and a few hours later the soil is still moist but the leaves have wilted on a hot, dry, windy afternoon. Reading the situation, the best explanation is that:

(A) The moist soil water has suddenly turned poisonous

(B) The leaves are losing water by transpiration faster than the roots can supply it

(C) The xylem has transformed into phloem and stopped carrying water

(D) The plant has permanently stopped making food

43. Water and dissolved minerals taken up from the soil must reach the leaves at the top of a tall plant. The tissue and direction responsible are:

(A) The phloem, carrying water and minerals upward from roots to leaves

(B) The xylem, carrying food downward from leaves to roots

(C) The stomata, drawing water sideways around the stem

(D) The xylem, carrying water and minerals upward from roots to leaves

44. Root hairs are tiny outgrowths on young roots. The feature that makes them so effective at taking up water and minerals is that they:

(A) Greatly increase the surface area of the root in contact with soil water

(B) Are green and make food to power the absorption

(C) Contain tiny muscles that actively pump water upward

(D) Carry food downward from the leaves into the soil

45. As water evaporates from the leaves through the stomata, a pull is set up that draws a continuous column of water up the stem. Naming this driving force and its origin:

(A) A heartbeat in the stem that pumps water up rhythmically

(B) A phloem push starting at the roots and shoving water upward

(C) The transpiration pull, created by water evaporating from the leaves

(D) A clotting pressure that thickens the sap and lifts it

46. Two identical leafy shoots stand in water. Shoot 1 is left in still, humid shade; shoot 2 is placed in hot, dry, moving air. Predicting the water uptake, shoot 2 will:

(A) Take up water more slowly, because heat seals the stomata shut

(B) Take up water faster, because hot dry moving air speeds transpiration and so increases the pull

(C) Take up exactly the same amount, since conditions cannot change transpiration

(D) Push water back down into the container, reversing the flow

47. If the xylem of a young stem were completely blocked while the phloem stayed intact, the part of the plant to suffer FIRST and why would be:

(A) The leaves, because food can no longer be carried up to them

(B) The roots, because food made above can no longer reach them

(C) The roots, because the minerals they absorbed cannot leave

(D) The leaves, because water can no longer be carried up to them

48. Food made in the leaves needs to reach a growing root tip below and a ripening fruit above at the same time. The tissue that makes this simultaneous two-way delivery possible is the:

(A) Xylem, which carries food in every direction

(B) Stomata, which push food out to the fruit and root

(C) Phloem, which can move food in more than one direction at once

(D) Root hairs, which pass food up to the leaves

49. A ring of bark, including the phloem, is stripped all the way around a tree trunk, leaving the inner wood untouched. Over time, the region that starves and the reason are:

(A) The leaves above the ring, because water can no longer rise to them

(B) The trunk and roots below the ring, because food made in the leaves can no longer travel down to them

(C) The whole tree at once, because the xylem was cut by the ring

(D) Only the bark on the ring, because that is the part removed

50. Which row correctly matches each transport tube with both its cargo and, where it applies, its direction?

(A) Xylem carries food downward; phloem carries water upward; ureter carries blood

(B) Xylem carries water upward; phloem carries water downward; ureter carries food

(C) Xylem carries urine; phloem carries minerals upward; ureter carries water

(D) Xylem carries water and minerals upward; phloem carries food in more than one direction; ureter carries urine

51. An Amoeba, a single cell, manages without any heart, vessels, or xylem, while a tall tree and a human cannot. The deciding factor is that the Amoeba:

(A) Does not need oxygen, food, or water to stay alive

(B) Is so small that materials can enter and leave directly across its surface by diffusion

(C) Keeps its blood inside one very large single cell

(D) Is too simple to require any movement of materials at all

52. A large tree and a human both need transport systems, yet a microscopic organism does not. The underlying reason a big multicellular body needs transport is that:

(A) It has no outer surface across which anything can pass

(B) It never requires oxygen, food, or water of any kind

(C) Its inner cells lie too far from the surface to be supplied by simple diffusion alone

(D) It can only survive while completely surrounded by water

53. Both the human kidney and a plant's transpiration centrally involve the movement of one substance. That shared substance is:

(A) Haemoglobin
(B) Platelets
(C) Bone
(D) Water

54. A student builds a comparison table of animal and plant transport. Which single analogy in the table is the most reasonable?

(A) The plant's xylem works like the veins returning blood to the heart

(B) The plant's phloem works like the kidneys filtering out waste

(C) The heart pumps blood through vessels in animals much as the transpiration pull draws water up the xylem in plants, except that plants have no pump

(D) A plant's root hairs act like the heart's pumping ventricles

55. A summary chart of transport in living things is being checked for the single fully correct row. That row is:

(A) In animals the xylem pumps blood; in plants the heart carries water

(B) In animals the veins carry blood away from the heart; in plants the phloem carries water up

(C) Both animals and plants use a heart to move all their materials

(D) In animals the heart pumps blood through vessels; in plants the xylem carries water up and the phloem carries food

56. A doctor records that a patient's wrist pulse and the 'lub-dub' counted with a stethoscope give the same number per minute. The reason these two counts match is that:

(A) The pulse is made by veins while the sound is made by arteries, by coincidence

(B) Sound travels faster than blood, so the numbers only seem equal

(C) The stethoscope counts breaths while the wrist counts beats, which happen to agree

(D) Each heartbeat both makes the valve sounds and sends one surge into the arteries felt as one pulse

57. Considering the whole circulation, why must blood keep moving continuously rather than being delivered once and left in the tissues?

(A) Because still blood would change from red to blue inside the body

(B) Cells constantly use up oxygen and food and keep producing waste, so blood must keep bringing supplies and carrying waste away

(C) Because the heart's valves would otherwise make no sound

(D) Because moving blood is the only thing that keeps it from becoming urine

58. A drop of blood is described by three facts: it is mostly a pale liquid by volume, it has by far the most numerous cells without nuclei, and it carries fragments that help seal cuts. Matching each fact to the correct component gives:

(A) The liquid is plasma, the numerous cells are white cells, and the sealing fragments are red cells

(B) The liquid is haemoglobin, the numerous cells are platelets, and the sealing fragments are white cells

(C) The liquid is plasma, the numerous nucleus-free cells are red cells, and the sealing fragments are platelets

(D) The liquid is urea, the numerous cells are red cells, and the sealing fragments are plasma

59. A model of the circulation uses a pump, thick stiff pipes, a fine mesh of thin tubing, and floppy valved return pipes. Mapping each model part to the real structure, the fine mesh of thin tubing represents the:

(A) Arteries, which carry blood away from the heart at high pressure

(B) Veins, which return blood to the heart through valves

(C) Capillaries, where exchange with body cells takes place

(D) Heart chambers, which collect and pump the blood

60. A gardener wants to slow water loss from a leafy plant during a heatwave without harming it. Reasoning from how water moves in plants, the change most likely to reduce water loss is to:

(A) Place a fan blowing warm dry air across the leaves to help it breathe

(B) Block the root hairs so the plant cannot pull up any water

(C) Seal the xylem so water can no longer reach the leaves at all

(D) Move the plant into still, humid shade, since cooler humid still air slows transpiration from the leaves

Solutions — Science (Class 7), Chapter 11: Transportation in Animals and Plants — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	A	21	A	31	B	41	A	51	B
2	D	12	D	22	B	32	A	42	B	52	C
3	A	13	C	23	D	33	C	43	D	53	D
4	A	14	D	24	C	34	D	44	A	54	C
5	A	15	A	25	A	35	B	45	C	55	D
6	D	16	B	26	A	36	B	46	B	56	D
7	C	17	B	27	C	37	B	47	D	57	B
8	A	18	C	28	B	38	D	48	C	58	C
9	C	19	B	29	D	39	A	49	B	59	C
10	C	20	B	30	A	40	C	50	D	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Normal oxygen capacity rules out a red-cell or haemoglobin problem, and normal clotting time rules out platelets; the only failing job is defence, which belongs to the white cells. Plasma carries dissolved substances and the cells but does not itself engulf germs.

2. (D) The pale-yellow upper column is plasma and the packed lower layer is the cells; a shorter yellow column with a thicker packed layer means proportionally less plasma and more cells. Haemoglobin sits inside the red cells, not dissolved in the plasma, so the upper layer's size is not a haemoglobin reading.

3. (A) Without a nucleus there is more room inside for the oxygen-binding pigment haemoglobin, which is the red cell's whole purpose; the cost is that such a cell cannot divide. Oxygen carriage depends on haemoglobin, not on having a nucleus, and clotting and defence are the jobs of platelets and white cells.

4. (A) Breathlessness from poor oxygen delivery traces to haemoglobin in red cells, while prolonged bleeding traces to platelets that seal wounds. White cells defend against germs, and plasma volume is unrelated to either symptom.

5. (A) White cells are the body's defence force, so their numbers rise to meet an active infection and ease off when it is over. The cell types do not convert into one another, and clotting (platelets) is not what fights an infection.

6. (D) Iron is a component of haemoglobin, the pigment that binds oxygen; too little iron means too little haemoglobin and reduced oxygen carriage. Clotting, the liquid medium, and defence depend on platelets, plasma, and white cells respectively, none of which is the iron story.

7. (C) Plasma is the liquid that moves dissolved digested food, wastes such as urea, and other materials between organs. Oxygen is mostly carried by haemoglobin in red cells, not by plasma, and food must be digested before it can travel dissolved in the blood.

8. (A) Oxygen carriage to tissues is the job of haemoglobin inside red cells, so they are the component that restores oxygen delivery. Clotting, defence, and the liquid medium do not themselves transport oxygen.

9. (C) Oxygen-rich blood is brighter red and oxygen-poor blood is darker red because the colour depends on how much oxygen is bound to haemoglobin. Platelet count does not set the colour, and blood is never blue, nor does it become urine.

10. (C) The most numerous, small, round, nucleus-free cells are the red cells, and their function is oxygen transport. White cells defend, platelets help clotting, and plasma is a liquid, not a cell type.

11. (A) Arteries receive blood directly from the pumping heart at high, pulsing pressure, so a cut artery spurts in time with each beat; veins return blood at low steady pressure and merely ooze. Valves are a vein feature, and arteries are the thick-walled vessels, not the thin ones.

12. (D) Thick muscular walls with no valves mark an artery built for high pressure; thin walls with valves mark a vein carrying low-pressure blood back to the heart; a wall one cell thick is the exchange capillary. Swapping any pair contradicts these structural rules.

13. (C) Blood in veins flows at low pressure, so without valves it could drift backward; the valves act as one-way gates keeping it moving toward the heart. The high-pressure claim is exactly backwards, and valves neither make heart sounds nor pump.

14. (D) Blood leaves the heart in arteries, reaches the tissue through thin capillaries where exchange happens, and returns in veins. Any order that puts capillaries first or swaps the artery and vein ends reverses the direction of flow.

15. (A) The rule of thumb generally holds, but the lung vessels invert it: the artery to the lungs is oxygen-poor and the vein from the lungs is oxygen-rich. The 'no exceptions' and 'always reversed' versions are both wrong, and it does not depend on age.

16. (B) Arteries take blood straight from the pump, so pressure peaks there, while the tiny capillaries slow the flow right down to allow exchange. Veins are low-pressure return vessels, not the high-pressure ones.

17. (B) Human blood is always red — bright when oxygen-rich, dark when oxygen-poor — and the bluish surface appearance is an effect of light through skin, not a real blue colour. Plasma is pale yellow, not blue.

18. (C) Arteries alone have the thick elastic muscular walls needed for high pressure; valves are a vein feature, one-cell-thin walls describe capillaries, and carrying blood toward the heart is the vein's role, not the artery's.

19. (B) The capillary's one-cell-thick wall is precisely what allows gases, food, and wastes to diffuse across; thickening it would block that exchange. Thick walls do not push substances out, capillaries do not pump, and wall thickness is unrelated to clotting.

20. (B) Carrying blood to the heart, having valves, and oozing at low pressure all identify a vein, and its valves prevent backward slippage of low-pressure blood. Arteries lack valves, capillaries have none either, and the 'valves for high pressure' reason is the reversed misconception.

21. (A) Separate sides stop oxygen-rich and oxygen-poor blood from diluting each other, so the body receives blood at full oxygen content. Loudness, colour-change speed, and muscle-free beating are not real consequences of the separation.

22. (B) The lower chambers, the ventricles, push blood out under force, so they need thicker muscle, while the upper atria mostly receive returning blood. The reversed receive/pump option, urine storage, and pulse-making are all wrong roles for these chambers.

23. (D) Each 'lub' and 'dub' is the sound of heart valves snapping shut during the pumping cycle. Capillary flow is silent at this scale, breathing is a separate sound, and the wrist throb is the pulse, not the heart sound.

24. (C) The pulse is the rhythmic stretching of an artery wall as each heartbeat sends a pressure wave of blood through it. It is not a valve snap, a clot, or lung air, and it occurs in arteries near the surface.

25. (A) In 18 seconds there are 27 beats, so the rate is 27 divided by 18, then multiplied by 60, giving 1.5 times 60, which is 90 beats per minute. Simply keeping 27, or doubling or under-scaling, mishandles the time conversion.

26. (A) Doubling the rate from 75 to 150 means twice as many beats in the same 20 seconds, which speeds delivery of oxygen and food and removal of waste to hard-working muscles. The half-as-often and unchanged options contradict the doubling, and clotting is not the reason.

27. (C) A resting rate of 132 is far above the typical resting value of about 72, so it counts as much higher than normal. It is neither normal nor low, and such a rate, while high, is not impossible for a living person.

28. (B) Routing returning blood through the lungs to pick up oxygen before it goes to the body ensures the body always receives oxygen-rich blood. Sending oxygen-poor blood straight to the body, skipping the lungs, or mixing the two would all reduce oxygen delivery.

29. (D) If one beat occurs every 0.8 seconds, the number per minute is 60 divided by 0.8, which is 75, a normal resting value. The other options come from dividing or multiplying incorrectly.

30. (A) The heart is rhythmically contracting muscle that pumps blood, not bone, and it neither filters wastes (the kidney's job) nor makes or ships food (and phloem belongs to plants, not animals).

31. (B) Water rises in the xylem because transpiration from the leaves creates a pull, doing the lifting job without any pump, which is the closest parallel to the heart. Plants have no heart, the phloem carries food not water, and stomata are leaf pores, not soil-suckers.

32. (A) Excretion is the removal of waste; taking in and digesting food is nutrition, the opposite of removing waste. Forming urine, sweating, and passing nitrogen waste through the kidneys are all genuine excretion.

33. (C) The kidney's filtering units are the nephrons, and they lie within the kidneys themselves. Stomata are leaf pores, platelets are blood components, and the filtering does not happen in the bladder, which only stores urine.

34. (D) Urine forms in the kidney, runs down the ureter to the bladder for storage, and leaves through the urethra. The other sequences misplace the urethra or start the path in the wrong organ.

35. (B) Urea is a waste made in body cells, travels dissolved in the blood, is filtered by the kidneys, and leaves in urine. The other paths invent plant tissues in a human, make urea in the wrong organ, or route it through the heart.

36. (B) Urea is a poisonous nitrogen waste that must be removed continually, not a food or a reserve, so storing it would harm the body. It does not nourish the body, colour the blood, or get banked in the kidneys.

37. (B) Both sweating and urine formation remove water and dissolved wastes, so on a hot day more leaves as sweat and less as urine, balancing the body's water. Sweating does not switch off the kidneys, sweat does carry some waste, and neither process takes in oxygen.

38. (D) Dialysis is the machine process that filters wastes from the blood when the kidneys cannot, doing the kidneys' work. A transfusion supplies blood, transpiration is a plant process, and circulation is the heart's normal pumping, none of which filters waste in kidney failure.

39. (A) Plants get rid of some gaseous waste and excess water through stomata and store other wastes in leaves, bark, gums, and resins. They have no kidneys, do not route all waste up the xylem to the roots, and do produce waste.

40. (C) Gums, resins, and waste-laden leaves that fall are routes by which plants store or shed waste. They are not food transport, water rising, or oxygen release, which are different processes.

41. (A) Plants form less harmful waste and can store much of it, so their excretion is slower and simpler than the active filtering animals do. They have no kidneys or heart for this, and they do produce waste.

42. (B) Hot, dry, windy conditions speed transpiration so the leaves lose water faster than the roots replace it, causing temporary wilting even in moist soil. Poisoned soil, tissue transformation, and a permanent halt to food-making are not what happens.

43. (D) Upward transport of water and minerals from roots to leaves is the xylem's job. The phloem carries food rather than the soil water, the xylem does not haul food, and stomata are leaf pores that do not move water through the stem.

44. (A) Many fine root hairs hugely enlarge the root's surface area touching soil water, boosting absorption. They are not green or food-making, have no muscles, and do not carry food into the soil.

45. (C) Evaporation of water from the leaves generates a suction, the transpiration pull, that lifts water up the xylem. Plants have no heart, the phloem carries food not water, and clotting is a blood process, not a plant lifting force.

46. (B) Hot, dry, moving air increases evaporation from the leaves, strengthening the transpiration pull, so shoot 2 draws up more water than the shoot in still humid shade. Heat does not seal the stomata, conditions do affect transpiration, and the flow does not reverse.

47. (D) The xylem lifts water to the leaves, so blocking it starves the leaves of water first; food is the phloem's cargo, which is still intact. The root-based options confuse the xylem's job with the phloem's.

48. (C) The phloem transports food and can carry it both up and down at the same time, reaching a root tip and a fruit together. The xylem moves water not food, stomata are gas pores, and root hairs absorb water, not deliver food.

49. (B) Removing the phloem ring breaks the downward food path, so the parts below — trunk and roots — are cut off from leaf-made food and starve, while water still rises in the untouched xylem to feed the leaves. The xylem was not cut, so water transport continues.

50. (D) The xylem moves water and minerals up, the phloem moves food in multiple directions, and the ureter carries urine. The other rows swap the xylem and phloem cargoes or misassign the ureter's load.

51. (B) Because it is tiny, every part of an Amoeba is close to its surface, so simple diffusion supplies and removes materials without a transport system. It still needs oxygen and food, has no blood, and does move materials — just by diffusion.

52. (C) In a large body the deep cells are far from the surface, so diffusion alone is too slow and a transport system is needed to carry supplies and wastes. Such bodies do have surfaces, do need oxygen and food, and are not limited to water.

53. (D) The kidney moves and adjusts water as it forms urine, and transpiration is the loss of water vapour from leaves, so water is common to both. Haemoglobin and platelets are blood items, and bone has no role in either process.

54. (C) The closest parallel is that the heart's pumping and the transpiration pull both drive a fluid through a system, with the caveat that plants lack an actual pump. Xylem moves water up rather than returning fluid like veins, phloem carries food rather than filtering waste, and root hairs absorb rather than pump.

55. (D) Animals rely on a pumping heart and blood vessels, while plants use the xylem for upward water and the phloem for food, which is the consistent picture. The other rows put plant tissues in animals, reverse the vein direction, or invent a plant heart.

56. (D) One heartbeat produces both the valve-closing sounds and a single pressure surge into the arteries, so the heart-sound count and the pulse count are the same event measured two ways. The pulse is not a vein effect, the match is not a coincidence of speeds, and the stethoscope is not counting breaths.

57. (B) Body cells continually consume oxygen and food and generate waste, so circulation must run nonstop to resupply and clear them. Blood does not turn blue when still, valve sounds are not the purpose of flow, and blood does not become urine.

58. (C) Plasma is the liquid bulk, red cells are the most numerous and lack a nucleus, and platelets are the fragments that seal wounds. White cells are far fewer and defend rather than seal, haemoglobin sits inside red cells rather than being the liquid, and urea is a dissolved waste, not the liquid medium.

59. (C) A fine mesh of very thin tubing matches the capillaries, whose one-cell-thick walls allow exchange. The thick stiff pipes are the arteries, the floppy valved return pipes are the veins, and the pump is the heart.

60. (D) Transpiration slows in cool, humid, still air, so shading the plant cuts water loss without harming it. Warm dry moving air would speed loss, and blocking root hairs or sealing the xylem would starve the plant of water rather than gently reduce loss.